



IEEE NITP
STUDENT BRANCH



HACK O' FEST

HACK O' FEST

HACK O' FEST

HACK O' FEST

HACK O' FEST

HACK O' FEST

HACK O' FEST

HACK O' FEST

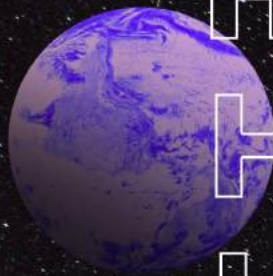
HACK O' FEST

HACK O' FEST

HACK O' FEST

HACK O' FEST

HACK O' FEST



RULEBOOK
noun
the regulation or standards of behaviour that should be followed in a particular job, organisation, or event





TIMELINE

Registration Starts	1st March 2024
Info Session	First week of March 2024
PPT Submission Start	5th March 2024
Inaugral Ceremony	1st April 2024
Registration Ends	2nd April 2024
PPT Submission Ends	4th April 2024
Speaker Session & Workshop	1st - 5th April 2024
Hackathon Starts	6th April 2024 14:00 IST
Hackathon Ends	7th April 2024 20:00 IST
Community review Window Starts	8th April 2024
Community review Window Ends	14th April 2024
Valedictory Ceremony	14th April 2024

THEME

Software Domain

- Quality Education
- Cyber Security
- Healthcare and Medical Facility
- Business and Economic Growth
- Sustainability
- Laws and Justice
- Community Building
- Space and Aerospace Technology
- Open Innovation

Hardware Domain

- Internet of Things (IoT) Solution
- Wearables Technology
- Robotics
- Home Automation
- Drone Technology
- Healthcare Innovation
- Green Technology
- AI-Hardware
- Security and Privacy

TRACKS

- Web Development
- Augmented Reality and Virtual Reality
- Cryptography and Network Security
- App Development
- Web3 - Blockchain
- Artificial Intelligence

SPECIAL TRACKS

- Web3/ Ethereum Track
- NLP toolkit challenge
- IoT (Internet of Things)
- Robotics
- Polygon Track
- Sentiment Analysis



**(Special privilege for the one making projects on Sentiment analysis, NLP toolkit challenge)

SIDE TRACKS

- Entrepreneurship Opportunity and Innovation

Web3/Ethereum Track Rules

Projects using blockchain technology, especially those utilizing the Ethereum network and decentralized apps (dApps), are the focus of this track. The biggest Web3 Hackathon in the nation. Before you participate, please carefully read the following rules and guidelines

PLATFORM REQUIREMENT & SMART CONTRACT DEVELOPMENT

- Submissions for the Web3 track must be developed on networks that are compatible with ethereum or the Ethereum blockchain.
- Using Solidity or the other Ethereum-compatible programming languages, participants are encouraged to create smart contracts.
- A Web3 wallet solution such as MetaMask shall be integrated by each team. Facilitates transactions and interactions with the Ethereum blockchain.

JUDGING CRITERIA

- Projects will undergo evaluation according to factors like innovation , technical intricacy, usability, and impact.
- Creating full-stack applications with smart contract integration is encouraged to demonstrate the extensive use of blockchain technology.

Entrepreneurship Ideas

This hackathon features a special "Incubation Track" focused on identifying promising startup ideas. Unlike regular tracks, this one emphasizes scalability, revenue potential, and overall business viability. The top 3 teams will win incubation at NIT Patna, including mentorship, resources, and connections to investors - everything to launch your dream startup!

SPECIAL SIDE TRACK:

Entrepreneurship Opportunity and Innovation
(Hardware Track Elaboration IoT & Robotics Technology)

At this hackathon, we're introducing a special track focused on evaluating projects with startup potential. Unlike other tracks, the Incubation Center will assess projects based on factors like scalability and revenue generation capability. The top three teams selected will have opportunity to be incubated at the NIT Patna Incubation Center. This platform offers budding entrepreneurs guidance, mentorship from industry experts, access to resources and connections with potential investors to nurture their business ideas.

SPECIAL SUPPORT TO BUDDING ENTREPRENEURS

******(Exclusively awarded to the winners)

The team with the most innovative startup idea will receive assistance in presenting their concept to the incubation center at NIT Patna. Moreover, they will be financially supported through reimbursement of their invested capital, fostering the development of their startup venture from conception to reality.

NLP Toolkit Challenge

The Natural Language Processing Toolkit (NLTK) challenge is an event aimed at testing participants' skills in utilizing NLTK, a popular Python library for natural language processing tasks such as tokenization, parsing, and sentiment analysis. Participants typically engage in tasks like text classification, named entity recognition, and machine translation using NLTK's functionalities.

REQUIREMENTS AND REQUISITES

- Participants should be proficient in a programming language commonly used in NLP, such as Python, Java, or C++.
- Require participants to document their approach, including details such as the algorithms used, any preprocessing steps, model architecture, hyperparameters, etc. This documentation should be clear and comprehensive to facilitate understanding and replication of their work.
- Toolkit framework that participants use includes NLTK (Natural Language Toolkit), spaCy, Stanford NLP, ALIENLP, Hugging Face Transformers, etc.

JUDGING CRITERIA

- **Solution Effectiveness:** Evaluate the performance of the NLP solution in terms of its accuracy, precision, recall, F1 score, BLEU score (for machine translation), ROUGE score (for text summarization), or any other relevant evaluation metric specific to the task(s) tackled.
- **Innovation:** Assess the novelty and creativity of the approach taken by the participants. This could include innovative use of the NLP toolkit, creative problem-solving, or novel combinations of techniques to address the challenge.
- **Scalability:** Consider the scalability of the solution, particularly its ability to handle large datasets or adapt to different domains or language. Scalability could be evaluated in terms of computational efficiency, memory usage, and performance under increasing workload.

SENTIMENT ANALYSIS

Sentiment analysis is a natural language processing (NLP) technique used to determine the sentiment or emotional tone expressed in a piece of text. It involves analyzing text data to categorize it as positive, negative, or neutral based on the underlying sentiment conveyed by the words and phrases used. Sentiment analysis algorithms often employ machine learning and statistical techniques to automatically classify text into sentiment categories, enabling businesses and organizations to gain insights into public opinion, customer feedback, social media discussion, and other textual data sources.

REQUIREMENTS AND REQUISITES

- Participants should be proficient in programming language commonly used in Natural Language Processing (NLP) tasks, such as Python, Java, or R.
- Participants should have a basic understanding of sentiment analysis concepts, including sentiment lexicons, machine learning algorithms for classification, and evaluation metrics.
- Participants need to solve tasks such as sentiment classification(binary or multiclass), aspect-based sentiment analysis, sentiment intensity prediction, or sentiment trend analysis.
- Evaluation metric(s) that participants should use to assess the performance of their sentiment analysis models include accuracy, precision, recall, F1 score, or area under the receiver operating characteristics curve (AUC-ROC).

PRESENTATION

- **Introduction (5 minutes):**
 - Introducing the team members.
 - Provide a brief overview of the sentiment analysis task tackled.
 - Summarize the approach taken to address the task.
- **Problem Statement (5 minutes):**
 - Clearly define the problem statement and its significance.
 - Describe the dataset(s) used for sentiment analysis.
 - Highlight any challenges or complexities encountered during the task.
- **Methodology (10 minutes):**
 - Explain the preprocessing steps applied to the dataset(s), including text cleaning, tokenization, normalization, and feature extraction.
 - Describe the sentiment analysis techniques or models employed, such as machine learning algorithms (e.g., Naive Bayes, Support Vector Machines, Deep learning models like LSTM) or lexicon-based methods.
 - Discuss any innovative or novel approaches utilizes in the sentiment analysis process.

• **Result and Analysis (10 minutes):**

- Present the experimental result obtained from the sentiment analysis models.
- Discuss the performance metrics achieved, such as accuracy, precision, recall, F1 score or other relevant evaluation metrics.
- Provide insights into the strengths and limitations of the developed sentiment analysis solution.
- Compare the performance of different models or techniques explored, if applicable.

• **Demo (10 minutes):**

- Demonstrate the functionality of the sentiment analysis solution in action.
- Showcase how the solution can be used to analyze sentiment in real world text data.
- Provide examples of input text samples and the corresponding sentiment predictions generated by the model.

INTERNET OF THINGS

The Internet of Things (IoT) refers to the network of physical devices, vehicles, appliances, and other items embedded with sensors, software, and connectivity, enabling them to connect and exchange data over the internet. Essentially, IoT involves the interconnection of everyday objects to the internet and each other, allowing them to collect, transmit, and receive data, as well as perform actions based on that data.

REQUIREMENTS AND REQUISITES

- Strong problem-solving skills and creativity are essential for participants to identify innovative IoT solutions that address real-world challenges effectively.
- Basic knowledge of electronics and hardware components, such as microcontrollers, sensors, actuators, and communication models, is necessary for participants to build and prototype IoT devices.
- Familiarity with the fundamentals of IoT, including sensors, actuators, communication protocols, edge computing, cloud computing, and IoT platforms, is essential for participants to develop effective IoT solutions.

JUDGING CRITERIA

- **Innovation and Creativity:** Assess the originality creativity of the IoT solution. Consider whether the solution introduces new ideas, utilizes emerging technologies, or addresses a unique problem innovatively.
- **Functionality and Performance:** Evaluate the functionality and performance of the IoT prototype. Consider factors such as reliability, responsiveness, scalability, and efficiency in performing its intended tasks
- **Impact and Practicality:** Assess the potential impact and practicality of the IoT solution in addressing real-world challenges or opportunities. Consider factors such as the relevance of the problem addressed, the potential for widespread adoption, and the positive social or economic implications of the solution.
- **Integration and Interoperability:** Assess the degree to which the IoT solution integrates with existing systems, platforms, or infrastructure, as well as its interoperability with other IoT devices or services,. Consider whether the solution follows open standards and protocols to facilitate interoperability and compatibility.

PRESENTATION

- **Introduction (5 minutes):**
 - Introducing the team members.
 - Provide a brief overview of the IoT project and its objectives.
- **Problem Statement (5 minutes):**
 - Clearly define the problem or challenge that the IoT solution aims to address.
 - Explain the significance and potential impact of solving the problem.
- **Software Architecture (10 minutes):**
 - Present the architecture of the IoT solution , including hardware components, sensors, actuators, communication protocols, and cloud services used.
 - Describe how data is collected, processed, and transmitted within the IoT system.
 - Explain any innovative or unique aspects of the solution architecture.
- **Implementation (15 minutes):**
 - Demonstrate the implementation of IoT solution, including the hardware prototype and software applications developed.
 - Showcase the integration of sensors, actuators, and IoT devices.
 - Provide insights into the development process, challenges faced, and solutions implemented

• **Implementation (15 minutes):**

- Demonstrate the implementation of IoT solution, including the hardware prototype and software applications developed.
- Showcase the integration of sensors, actuators, and IoT devices.
- Provide insights into the development process, challenges faced, and solutions implemented

• **Functionality and Features (10minutes):**

- Highlight the key functionalities and features of the IoT solution.
- Showcase how the solution addresses the problem statement and fulfills the project objectives.
- Provide examples or scenarios demonstrating the use of the IoT solution in real-world applications.

• **Data Analysis and Insights (10minutes):**

- Discuss how data collected from IoT devices is analyzed and interpreted to derive meaningful insights.
- Present any data visualization techniques or analytics algorithms used to extract insights from the data.
- Share any interesting findings or observation obtained from the data analysis.

TERMS AND CONDITIONS OF PARTICIPATION

- Before the hackathon , teams will participate in online project idea submission in form of presentation.
- The submitted idea presentations must adhere to a provided PowerPoint template, which will be distributed later via Devfolio's Announcement Section.
- A maximum of 25-30 teams will be selected for the offline round based on their ideas.
- Evaluation of project overviews will be conducted by the faculty advisor committee of Hack O' Fest from NIT Patna and a panel of expert judges.
- All teams, except those shortlisted for the offline round in Patna, will engage in the hackathon exclusively through online participation.
- It is emphasized that all teams will undergo evaluation based on standardized criteria for their hackathon projects, ensuring impartial assessment across both offline and online participants.

COMMUNICATION AND DISCORD

- Each member of a team must join the Hack O' Fest Discord Server before the hackathon starts. Also, every member must change their server nickname to their official name.
- Each team must open a Support ticket on the Hack O' Fest Discord Server and get their dedicated private team channel. Failing to do so will lead to the cancellation of participation of the team from the event.
- All the announcements about the event will be made on the Hack O' Fest Discord Server. Hereby, it is advisable to keep your notifications on for the server and be active to not miss out on any important announcements.
- Teams can seek help/support from the organizers via the public channel or their private channel.
- No participant should personally contact or relay a direct message to any of the organizers, mentors, or judges. Failing to comply with the same, shall lead to immediate and unobjectionable disqualification.

CODE OF CONDUCT

- Participants must agree to adhere to the Hack O' Fest Community Guidelines and Code of Conduct upon joining the event, in addition to all event terms and conditions.
- Respectful treatment of all event participants, including mentors, judges, and organizers, is expected, aligning with the IEEE SB NITP and Think India SC NITP Community Guidelines and Code of Conduct.
- Organizers retain the right to disqualify teams from the event at their discretion, based on violations such as rule breaches, misconduct, or unprofessional behaviour.
- Any violation of terms and conditions by a team member will result in the entire team's disqualification from Hack O' Fest, with organizers' decisions being final.
- The organizing teams may adjust event terms and conditions at any time, participants' agreement to the initial terms is considered as acceptance of subsequent changes.
- Attendance at the opening ceremony is mandatory for all registered candidates, as vital guidelines will be presented for both online and offline mode.
- Each team must first select at least one track and then choose corresponding themes accordingly.
- Project work must be strictly confined to the 30-hour duration of the hackathon.
- Teams are permitted to utilize libraries, frameworks, or open-source code for project implementation.
- Pre-event project work and open sourcing solely for use during the event are prohibited, aligning with the spirit of the rules.
- While encouraged to submit projects developed during the hackathon, participants disclosing reused code segments or resubmitting previous projects must provide details about prior use.
- Projects found to reuse undisclosed code upon inspection may be subject to disqualification or require participants to differentiate between old and new work.
- Plagiarism or inclusion of explicit content in hacks will result in automatic disqualification, with organizers' decisions being final.

SUBMISSION GUIDELINES

RESPOSITORY

Software Domain:

- Create a new GitHub repository after the hackathon begins.
- Set the GitHub repository of the project to Public.
- Commit the project's code and resources into the repository.
- Organize the repository with clear folder structures.
- If possible, generate and store executables (e.g., Android APKs or JARs) in the repository.
- Ensure the GitHub repository remains unchanged and publicly accessible from the start of the hackathon until the results are declared.

Hardware Domain:

- The judging panel will be running an offline assessment while judging the projects and marking them accordingly.
- Proper and well-formatted documentation is required to be submitted via Google form circulated citing details of various technical aspects and functionalities of the project.

README FILE

- Include links to the Presentation and Video pitch in the README file.
- If applicable, add deployment links at the top of the README file.
- Provide setup instructions, if necessary, for running the project in the README file.
- List all project dependencies in the README file.
- Ensure the README file contains a clear description of how the hack works

CRUX OF JUDGING AND EVALUATION

ORIGINALITY OF IDEA

- The concept or idea's uniqueness and originality will be evaluated.
- Assessment will be made on how effectively the idea tackles the identified challenge by the team.
- The solution's potential impact will be considered.

FUNCTIONALITY

- Evaluation of the effectiveness of the solution's functionality.
- Assessment of the prototype or final product's completeness and reliability.

USER EXPERIENCE (UX)

- Consideration of the intuitiveness and user-friendliness for end-users.
- Evaluation of the design thoughtfulness and user interface quality.

ADAPTABILITY

- Assessment of the solution's ability to accommodate future changes or enhancements.

CODE QUALITY

- Evaluation of the cleanliness, organization, and readability of the codebase.
- Assessment of the implementation's adherence to coding best practices for optimal effectiveness.

PRESENTATION

- Assessment of the clarity and effectiveness of the presentation delivery.
- Evaluation of the team's ability to articulate the problem, proposed solution, and value proposition.

TECHNICAL COMPLEXITY & IMPLEMENTATION

- Review of technical challenges successfully resolved.
- Assessment of the complexity and sophistication demonstrated by chosen technology stack.

SCALABILITY

- The potential for the solution to scale to handle increased data, users, or usage.

FEASIBILITY

- Realistic and practical implementation within given constraints.
- Consideration of potential challenges and mitigation strategies.

IMPACT AND SOCIAL RELEVANCE

- Evaluation of the potential positive impact on users, communities, or society.
- Assessment of the solution's ability to effectively address real world problem or needs.

JUDGES' IMPRESSION

- Overall impression of the judges based on the team's passion, dedication, and effort through the MID-EVALUATION.

MID-EVALUATION

- The mid-evaluation will focus on gauging the advancements achieved in the initial phase of the hackathon.
- Judges will scrutinize task completion, code quality, and overall project progress up to the halfway mark.
- Teams must ensure a minimum number of significant commits throughout the hackathon duration
- Substantial progress in the project should be evident during the mid-evaluation.
- Teams should anticipate giving succinct and coherent responses during the mid-evaluation session.

DATE & TIME OF REGISTRATION

- The registration starts on MARCH 1, 2024 from 09:00 Hrs. (30 Hrs long offline event).
- There is no registration fee for participation.
- Registration for participants is invited via Devfolio.
- The timeline of events is from 1 to 7 April.
- The offline hackathon is scheduled to take place on the 6th and 7th of April.



IMPORTANT LINKS

REGISTRATION : [HACK-O-FEST | DEVFOLIO](#)

DISCORD : [HACK-O-FEST | DISCORD](#)

SOCIAL MEDIA HANDLES: [HACKOFEST NITP \(@HACKOFEST_NITP\)](#)